

SUMMARY

Platform and reliability engineer with 12+ years of experience designing and scaling mission-critical systems across Kubernetes, cloud infrastructure, and Linux. I lead cross-functional initiatives that improve reliability, security, and developer productivity across the organization. I design and build platforms and AI-enabled automation that cut operational toil, accelerate vulnerability remediation, enable zero-trust adoption, and help engineering teams ship faster and more safely.

SKILLS

Languages: Go, Python, Bash

Orchestration & Platform: Kubernetes, Helm, ArgoCD, Flux, Backstage

Cloud & IaC: AWS, GCP, Terraform, Ansible, Packer

Observability: Prometheus, Grafana, OpenTelemetry, ELK

CI/CD & Security: Jenkins, GitHub Actions, Semaphore, Atlantis, Vault, SPIRE, OPA

EXPERIENCE

- **Confluent** Remote
Senior Software Engineer II, Developer Productivity *April 2025 - Present*
 - **Vulnerability remediation:** Designed and shipped an AI agent that automates the full software-vulnerability remediation lifecycle; now the organization's default, it has remediated 300+ vulnerabilities in about five months. Championed its evolution onto a new autonomous-workflow platform, driving cross-team design reviews and coaching its owner to lead it.
 - **Developer platform:** Co-leading the technical direction of the internal developer platform, owning the developer-experience and self-service roadmap that helps engineering teams ship services faster and more safely.
 - **AI gateway:** Leading the selection and architecture of the company's AI gateway (the control plane for how services access large language models), where source-level verification caught a supply-chain compromise in a leading candidate before adoption. An earlier AI-infrastructure evaluation I led avoided over half a million dollars a year by recommending against an unready deployment.
 - **Zero-trust & secrets:** Drove fleet-wide security and reliability hardening, automating a zero-trust (mTLS) service migration with AI-guided tooling and moving services onto modern least-privilege secret management, reducing operational toil and shrinking the credential attack surface.
 - **Production reliability:** Re-architected a production policy-enforcement service that silently dropped critical configuration-change approvals whenever slow synchronous processing exceeded an upstream timeout. Moving it to an asynchronous, back-pressured design with dashboards and alerting eliminated a class of silent failures and the on-call toil they caused.
 - **Technical leadership & enablement:** Authored RFCs and one-pagers that steered cross-team AI-tooling decisions, taught engineers across the org to build high-quality AI-assistant plugins, and published several to the company's internal marketplace.
- **Angi Inc** Remote
Senior Infrastructure Engineer *Sept 2023 - Mar 2025*
 - **Angi One:**
 - Collaborated with cross-functional teams to build key components of the Angi One platform using Kubernetes controllers, Flux, and ArgoCD, enabling single-manifest deployments for applications and cloud services, reducing developer overhead and cutting production time by 50%.
 - Led efforts to automate documentation workflows based on CRDs, ensuring up-to-date documentation with every release, reducing manual effort and support inquiries.
 - Partnering with development and observability teams to enhance and automate component generation in the Backstage catalog, streamlining the onboarding process and improving system visibility.
- **Workday** Victoria, BC, Canada
Senior Software Development Engineer, DevOps *May 2023 - Sept 2023.*
Software Development Engineer III, DevOps *May 2021 - April 2023.*
Software Development Engineer II, DevOps *May 2019 - April 2021.*

- **Development and Operations:**
 - Implemented policy-based infrastructure management using OPA and Atlantis improving developer experience while providing them more control over their resources and helping them shift security best practices to the left.
 - Mentored developers and junior engineers as they grew into their role.
 - Created an automated AWS deployment pipeline to deploy microservice application cross account and cross region reducing operations toil by over 4 hours every week.
 - Actively manage, improve, and monitor infrastructure resources in DC and on AWS including but not limited to EC2, ECS, Route53, S3, RDS, Lambda, ES etc. using tools like ansible, terraform, wavefront, ELK stack and slackbot.
 - Writing Jenkinsfiles and improving shared libraries for automated build and deployment of several applications and services using Jenkins.
 - Migrating applications and services to an in-house flavor of Kubernetes platform.
- **Knowledge Share:** Spearheaded an initiative between operations and multiple dev teams to empower the developers with the knowledge they would need to take ownership of their services.
- **Scrum Master:** Helped team self organize and self manage by incorporating servant leadership principles.
- **NYU CUSP** Brooklyn, NY, USA
Associate Research Scientist *May 2015 - May 2019*
Assistant Research Scientist *June 2014 - May 2015.*
 - **Dockkeeper:** Developed a scalable and secure container scheduling and monitoring tool that leverages the docker ecosystem and prometheus for provisioning services on physical hosts. This helped eliminate the VM license fees of over \$35,000 per year and optimizing the efficiency of hosts by over 55%.
 - **UOInfra:** Architected the NYU/CUSP Urban Observatory's multi-site physical infrastructure consisting of multiple dense compute and storage nodes comprising of over half a petabyte of storage space, provisioned for multi-user mini-HPC environments, using Ansible and Packer.
 Deployed a 27 screen vizwall using a cluster of networked raspberry Pi's enabling researchers to interpret their visualizable data, while keeping the whole price to 1/8 th of that of a commercial solution.
 - **SONYC:** Developed a secure machine critical IoT platform and implemented CI/CD framework for deploying and maintaining over 100 urban noise monitoring sensors in NYC. This project has won the \$ 4.6 Million CPS frontier award from NSF.

EDUCATION

- **NYU Tandon School of Engineering** Brooklyn, NY, USA
Master of Science in Telecommunication Networks *Aug. 2012 - May. 2014*
Thesis - CitySynth: Imaging with a Network of Devices

CERTIFICATIONS

- **CKA Certified Kubernetes Administrator:**

PROJECTS

- **AdguardController:** Developed a Kubernetes controller to automate DNS record management, significantly improving deployment efficiency and reducing manual errors.
- **UCSLHUB:** Developed a resilient and scalable back-end infrastructure using Kubernetes, jupyterhub and keycloak for hosting the CUSP's UCSL bootcamp **which is accessed by hundreds of students every year.**
- **MUON:** Developed a research and analysis platform for scientists in the Multi-City Urban Observatory Network at the University of Delaware.
- **Homelab:** Running a HCI homelab with ansible and terraform managed KVM nodes with Ceph running behind a Pfense firewall that is used for testing and development including emulation of distributed datacenters.

PUBLICATIONS, TEACHING EXPERIENCE AND MORE..

- **View it online:** <https://sharmamohit.com/>